How to Run the Mental Wellness Chatbot Project

This guide explains how to set up the project, run both the backend and chatbot, and understand the purpose of each file.

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1. Extract the Project Folder

1. Extract the ZIP file anywhere on your computer.
   Example folder structure:

C:\MentalHealthProject\project-code

2. Before continuing, delete the following folders if they appear:

- backend\venv
- chatbot\venv
- Any folder named **pycache**

Each teammate must recreate their own virtual environments.

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2. Install Python

Ensure that Python 3.10 or newer is installed.

Check the version:

python --version

If Python is missing, download it from:
https://www.python.org/downloads/

During installation, make sure to enable:
Add Python to PATH

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3. Open the Project in VS Code

1. Open Visual Studio Code
2. Select: File → Open Folder
3. Choose the extracted folder:

C:\MentalHealthProject\project-code

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4. Create Virtual Environments

The project has two independent components:

- **Backend (Flask server)**
- **Chatbot (Gradio interface)**

Each one must have **its own virtual environment**.

4.1 Create the Backend Environment

Open the VS Code terminal and switch to the backend folder:

cd backend

Create the virtual environment:

python -m venv venv

Activate it:

venv\Scripts\activate

You will now see:

(venv) C:\MentalHealthProject\project-code\backend>

4.2 Create the Chatbot Environment

Open a **new separate terminal** in VS Code.

Switch to the chatbot folder:

cd chatbot

Create the virtual environment:

python -m venv venv

Activate it:

venv\Scripts\activate

You will now see:

(venv) C:\MentalHealthProject\project-code\chatbot>

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5. Install Required Packages

Backend Packages

Inside the backend terminal (the one where the venv is active):

pip install -r requirements.txt

If the backend has no requirements file, install manually:

pip install flask
pip install tenseal
pip install python-dotenv
pip install requests

Chatbot Packages

Inside the chatbot terminal (its own venv):

pip install -r requirements.txt

If no requirements file exists, install manually:

pip install gradio==4.44.0
pip install huggingface_hub
pip install python-dotenv
pip install requests

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6. Configure the HuggingFace Token

Inside the **chatbot** folder, open the file:

chatbot.env

The file should contain:

HUGGINGFACE_API_TOKEN=your_token_here
HF_MODEL_ID=meta-llama/Llama-3.1-8B-Instruct

Notes:

- It is okay if all teammates use the same token **only if the owner allows it**.
- Otherwise, each teammate should generate their own HuggingFace API token.

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7. Start the Backend Server

The backend handles:

- Receiving encrypted answers
- Performing homomorphic encryption computations
- Returning encrypted results to the chatbot

Using the backend terminal:

```
cd backend
venv\Scripts\activate
python server.py
```

You should see:

```
Server is running (HE backend)
Running on http://127.0.0.1:5000
```

Leave this terminal running.

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8. Start the Chatbot Interface

Open the separate chatbot terminal.

Run:

```
cd chatbot
venv\Scripts\activate
python app.py
```

You will see:

Running on local URL: http://127.0.0.1:7860

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9. Open the Chatbot in a Browser

Visit:

http://127.0.0.1:7860

You can now use:

- The **Chat** tab (LLM emotional support assistant)
- The **Private Stress Check** tab (fully encrypted HE scoring)

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10. File Overview

Below is an explanation of the important files your teammates will see.

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backend Folder

File / Folder	Purpose
server.py	Receives encrypted answers, performs the homomorphic weighted-sum computation, and returns encrypted scores.
encryption_module.py	Implements CKKS context creation, encryption, decryption, serialization, and deserialization. Shared logic between client and server.
requirements.txt	Python dependencies for backend.
test_client.py	Optional tool to test the backend manually. Not needed for running the chatbot. Can be deleted.
venv/	Should be deleted before sharing the project.
pycache/	Temporary auto-generated files. Safe to delete.

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chatbot Folder

File / Folder	Purpose
app.py	Main Gradio UI. Handles chatbot logic, encrypted stress check, and calls the backend.
.env	Stores HuggingFace API token and model ID. Must be created or updated.
encryption_module.py	A clean copy of the backend encryption module, used client-side.
requirements.txt	Python dependencies for chatbot.
venv/	Delete before sharing. Each teammate creates their own.
pycache/	Can be deleted safely.

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11. Files That Should Be Deleted Before Sharing the ZIP

File / Folder	Reason
backend\venv	Every teammate must recreate it.
chatbot\venv	Every teammate must recreate it.
Any **pycache** folder	Temporary files.